Use:

Describe steps used to get the *B. canis* genomes from the 2024 Ukraine fastq files.

Tools used:

- versions:
  - other peoples programs
    - raven: 1.8.3
      - https://github.com/lbcb-sci/raven
    - minimap2 2.28-r1209
      - https://github.com/lh3/minimap2
    - GrapeTree 1.5.0
      - https://github.com/achtman-lab/GrapeTree
  - my bioTools programs (pre-dates global bioTools version):
    - filtsam 2024-05-14
    - tbCon 2024-08-07
    - rmHomo 2025-07-28

Citations

The `references.bibtex` file has all references in bibtex format. The only exception is Phylo.io veiwer, which I could not find a reference for.

Phylo tree

Could not find citation.

https://beta.phylo.io/viewer

raven

Vaser, R., Šikić, M. Time- and memory-efficient genome assembly with Raven. Nat Comput Sci 1, 332–336 (2021). https://doi.org/10.1038/s43588-021-00073-4

minimap2

Li, H. (2018). Minimap2: pairwise alignment for nucleotide sequences. Bioinformatics, 34:3094-3100. doi:10.1093/bioinformatics/bty191

Li, H. (2021). New strategies to improve minimap2 alignment accuracy. Bioinformatics, 37:4572-4574. doi:10.1093/bioinformatics/btab705

Grapetree

Z Zhou, NF Alikhan, MJ Sergeant, N Luhmann, C Vaz, AP Francisco, JA Carrico, M Achtman (2018) "GrapeTree: Visualization of core genomic relationships among 100,000 bacterial pathogens", Genome Res; doi: https://doi.org/10.1101/gr.232397.117

Rimsphere database

Publication: Abdel-Glil MY, Thomas P, Brandt C, Melzer F, Subbaiyan A, Chaudhuri P, Harmsen D, Jolley KA, Janowicz A, Garofolo G, Neubauer H, and Pletz MW. Core Genome Multilocus Sequence Typing Scheme for Improved Characterization and Epidemiological Surveillance of Pathogenic Brucella. J Clin Microbiol. 2022, 60: e0031122 [PubMed 35852343]

sequencing notes

- Run: BG_EBS_0924
  - Date: 20240919
  - ONT Library Kit: RBK114.24 V14 chem RB Rapid Barcoding Kit

02 assembly

The original fastq files came from two separate runs. They were mereged with `cat` and then uncompressed with `gzip`. The data on the SRA is the merged data. The data was saved in the input directory, for the rest of this document we will refere to it as `01-input/merged-reads.fq`.

Raven was used to assemble the genomes.

```
raven \
    data_from_SRA.fastq \
  > 02-raven/02-BG_EMBS-bar08_11.fa;
```

03 polishing

For a quick polishing we did two rounds with tbCon for chromosome 1 and one round of polshing for chromosome 2.

We really did 5 rounds, but found that chromsome one was best with two rounds, while chromosome two was best with one round.

We also identified which chromosomes the contigs belong to before this step. However, for clarity this is shown in the next step.

first chromosome

```
minimap2 \
    -a \
    <(head -n 2 02-raven/02-BG_EMBS-bar08_11.fa) \
    01-input/merged-reads.fq |
  tbCon \
    -sam - \
    -perc-snp-sup 0.3 \
    -min-depth 5 \
    -min-q 3 \
    -min-q-ins 3 |
  filtsam \
     -out-fasta \
     -sam - |
  sed \
     "1s/.*/>bar08_11-chrom01-delete/;" \
    > del.fasta;

minimap2 \
    -a \
    del.fasta \
    01-input/merged-reads.fq |
  tbCon \
    -sam - \
    -perc-snp-sup 0.3 \
    -min-depth 5 \
    -min-q 3 \
    -min-q-ins 3 |
  filtsam -out-fasta -sam - |
  sed \
     "1s/.*/>bar08_11-chrom01-polish/;" \
    > 03-polishing/03-BG_EMBS-bar08_11-chrom1-polish.fa;

rm del.fasta;
```

second chromosome

```
minimap2 \
    -a \
    <(tail -n 2 02-raven/02-BG_EMBS-bar08_11.fa) \
    01-input/merged-reads.fq |
  tbCon \
    -sam - \
    -perc-snp-sup 0.3 \
    -min-depth 5 \
```

```
    -min-q 3 \
    -min-q-ins 3 |
  filtsam -out-fasta -sam - |
  sed \
     "1s/.*/>bar08_11-chrom02-polish/;" \
    > 03-polishing/03-BG_EMBS-bar08_11-chrom1-polish.fa;
```

04 curation

We wanted to curate the genomes before further analysis, so we found a close hit on blast. We then used the reference and `rmHomo -homo 1` to remove indels smaller than 4 bases long.

find reference genome

The genomes from Raven were large contigs, so I split the genomes into roughly 1 megabase pair chunks and used blast to identify the host species for each. In all cases *B. canis* was present.

I downloaded a *B. canis* genome (CP016977) from Genbank to act as a reference. I split chromosone one (01-input/CP016977-B_canis_chrom1.fa) and chromosome two (01-input/CP016978-B_canis_chrom2.fa) into separate files. To prevent mapping errors when mapping the assembly to the reference I circularized both chromosomes using `awk`.

```
awk \
    '
      { # MAIN
         if(NR == 1)
            headStr = $0;
         else
            seqStr = seqStr $0;
      }; # MAIN

      END{printf "%s\n%s%s\n", headStr, seqStr, seqStr;};
    ' \
    01-input/CP016977-B_canis-chrom1.fa \
  > 01-input/CP016977-B_canis-chrom1-circ.fa;

awk \
    '
      { # MAIN
         if(NR == 1)
            headStr = $0;
         else
            seqStr = seqStr $0;
```

```
    }; # MAIN

    END{printf "%s\n%s%s\n", headStr, seqStr, seqStr;};
  ' \
  01-input/CP016978-B_canis-chrom2.fa \
> 01-input/CP016978-B_canis-chrom2-circ.fa;
```

chromosome 1

```
minimap2 \
    -a \
    --eqx \
    01-input/CP016977-B_canis-chrom1-circ.fa \
    03-polishing/03-BG_EMBS-bar08_11-chrom1-polish.fa |
  rmHomo \
    -homo 1 \
    -ref 01-input/CP016977-B_canis-chrom1-circ.fa |
  filtsam \
    -out-fasta \
    -sam - \
    -out 04-curation/04-BG_EMBS-bar08_11-chrom1-polish-rmHomo.fa;
```

chromosome 2

```
minimap2 \
    -a \
    --eqx \
    01-input/CP016978-B_canis-chrom2-circ.fa \
    03-polishing/03-BG_EMBS-bar08_11-chrom2-polish.fa |
  rmHomo \
    -homo 1 \
    -ref 01-input/CP016978-B_canis-chrom2-circ.fa |
  filtsam \
    -out-fasta \
    -sam - \
    -out 04-curation/04-BG_EMBS-bar08_11-chrom2-polish-rmHomo.fa;
```

04 notes

For both chromosome 1 and 2 we noticed several large indels remained, including a six G insert in a 10 base long G homopolymer in chromsome two. The other indels in chromosome one and two were in VNTR regions and so, might affect lineage calls.

05 database

We needed to build up a cgMLST database before we could run a cgMLST analysis. So, we downloaded as many *B. canis* full reference genomes from Genbank as possible (see 01-input/01-references.acc for accession numbers).

Then we used in house scripts to convert the reference genomes into cgMLST lineages.

getting cgMLST genes

The *Brucella spp.* version 1.0 cgMLST genes were downloaded from rimshpere https://www.cgmlst.org/ncs/schema/Brucella2114/.

The database was extracted using `unzip`.

```
mkdir 01-input/rimsphere;
cd 01-input/rimsphere;
mv <rimshpere_cgMLST_database>.zip rimshpere;
unzip <rimshpere_cgMLST_database>.zip;
cd ../../;
```

The database was then converted to my cgMLST scripts format using in house scripts.

```
sh 00-scripts/rimSphCgMlstCnvt.sh \
    01-input/01-B_canis-cgMLS-db \
    01-input/rimsphere/*.fasta;
rm -r "rimsphere";
```

The final names were `01-B_canis-cgMLST-db--lenDb.tsv` and `01-B_canis-cgMLST-db--seqDb.fa`.

The sequence database `01-B_canis-cgMLST-db--seqDb.fa` was compressed using `gzip` to avoid the github large upload size limit.

building the cgMLST reference database

The cgMLST reference database for our in house scripts was built using in house scripts and the rimsphere cgMLST database. The converted reference database was named `05-linRef/0t-linDb.tsv`.

```
firstBl="TRUE";

for strFa in 01-input/01-Bcanis-refs/*.fa;
do   # Loop: add references to database
   accStr="$(\
      basename "$strFa" | sed 's/Bcanis-//; s/.fa//;' \
   )";

   if [ ! -d "05-canisRefs/05-$accStr" ];
```

```
   then
      mkdir "05-canisRefs/05-$accStr";
   fi

   sh 00-scripts/cgMLST.sh \
       "$strFa" \
       "05-refLin/05-$accStr/05-$accStr" \
       01-input/01-B_canis-cgMLST-db--seqDb.fa.gz \
       01-input/01-B_canis-cgMLST-db--lenDb.tsv \
       $accStr;

   if [ "$firstBl" = "TRUE" ];
   then
      cat \
          "05-refLin/05-$accStr/05-$accStr-cgMLST-map-lin-sheet.tsv" \
        > "05-refLin/05-linDb.tsv";
      firstBl="FALSE";
   else
      tail -n+2 \
          "05-refLin/05-$accStr/05-$accStr-cgMLST-map-lin-sheet.tsv" \
        >> "05-refLin/05-linDb.tsv";
   fi
done # Loop: add references to database
```

06 adding sequence to database

The Ukraine sequence was added to the database and the closet

```
cat 04-curation/*.fa > del.fa;
sh 00-scripts/cgMLST.sh \
    del.fa \
    06-ukrLin/06-BG_EMBS-bar08_11-polish-rmHomo-cgMLST \
    01-input/01-B_canis-cgMLST-db--seqDb.fa.gz \
    01-input/01-B_canis-cgMLST-db--lenDb.tsv;
rm del.fa
```

I checked to make sure the four base G homopolymer had little effect using awk. None of the lineages detected had more then a one base insertion. I also checked the problem lineages and no lineage had an insertion or deletion.

```
awk '{if($6 > 0) print $0;};' \
   06-ukrLin/06-BG_EMBS-bar08_11-polish-rmHomo-cgMLST-cgMLST-map.tsv;
```

I then merged the reference and lineage databases with `cat` to get one lineage database.

```
cat \
```

```
    05-linDb.tsv \
    06-ukrLin/06-BG_EMBS-bar08_11-polish-rmHomo-cgMLST-cgMLST-map-lin-sheet.tsv \
  > 06-ukrLin/06-linDb-UKR.tsv;
```

07 cgMLST tree

I made a minimum spanning cgMLST using the reference database and grapetree. The output `.newick` was converted to an svg image using `phylo.io` https://beta.phylo.io/viewer. The svg image was then converted to a tiff image using `inkscape`.

First I needed to prepare the database for grape tree.

```
sed \
   '1s/^/#/; s/\*/0/g;' \
   06-ukrLin/06-linDb-UKR.tsv \
  > 07-tree/07-linDb-UKR-set.tsv;
```

Need to install grape tree by virtual envrioment, so to activate it I did `. ~/files/pythonEnv/bin/activate`.

```
grapetree \
    --profile 07-tree/07-linDb-UKR-set.tsv \
    --matrix MSTreeV2 \
  > 07-tree/07-linDb-UKR-set-grapTree.newick
```

08 gene extaction

get UKR sequence genes

I then extracted the genes useing an in house script and concatinated them together using `cat`.

```
cat 04-curation/*.fa > del.fa;
sh 00-scripts/extractGenes.sh \
    del.fa \
    01-input/01-B_canis-cgMLST-db--seqDb.fa.gz;
rm del.fa
mv \
    coordinates.tsv \
    08-extract/08-BG-EMBS-bar08_11-geneCoords.tsv
mv genes.fa 08-extract/08-BG-EMBS-bar08_11-genes.fa;
```

get reference genes

Next I needed to get the cgMLST genes in the references and cocatinate all genes into sequences. I included a *B. suis* as an outgroup.

```
for strFa in ./01-input/01-Bcanis-refs/*.fa;
do
   bash 00-scripts/extractGenes.sh \
       "$strFa" \
       01-input/01-B_canis-cgMLST-db--seqDb.fa.gz &&
     awk \
       -v idStr="$(\
           basename "$strFa" |
           sed 's/Bcanis-//; s/\.fa//;' \
         )" \
       '{
           if($0 ~ /^>/ || $0 ~ /^$/)
              next;
           seqStr = seqStr $0;
         }
         END{printf ">%s\n%s\n", idStr, seqStr;};
       ' genes.fa  \
  >> 08-extract/08-Bcanis-concate.fa &&
  rm genes.fa coordinates.tsv;
done
```

get outgroup genes

Then I needed to add in the concatinated ukraine sequence to the database.

```
awk \
    -v idStr="UKR-2024" \
    '{
        if($0 ~ /^>/ || $0 ~ /^$/)
           next;
        seqStr = seqStr $0;
      }
      END{printf ">%s\n%s\n", idStr, seqStr;};
    ' 08-extract/08-BG-EMBS-bar08_11-genes.fa \
  >> 08-extract/08-Bcanis-concate.fa;
```

Then I added the concatinated outgroup sequence

```
bash 00-scripts/extractGenes.sh \
    01-canis-input/Bsuis-CP000911.fa \
    01-canis-input/01-B_canis-cgMLST-db--seqDb.fa &&
  awk \
    -v idStr="Bsuis-CP000911" \
    '{
        if($0 ~ /^>/ || $0 ~ /^$/)
           next;
        seqStr = seqStr $0;
```

```
      }
      END{printf ">%s\n%s\n", idStr, seqStr;};
    ' genes.fa  \
  >> 08-extract/08-Bcanis-concate.fa &&
  rm genes.fa coordinates.tsv;
```

At this point the genomes should be ready for alignment and tree building.